

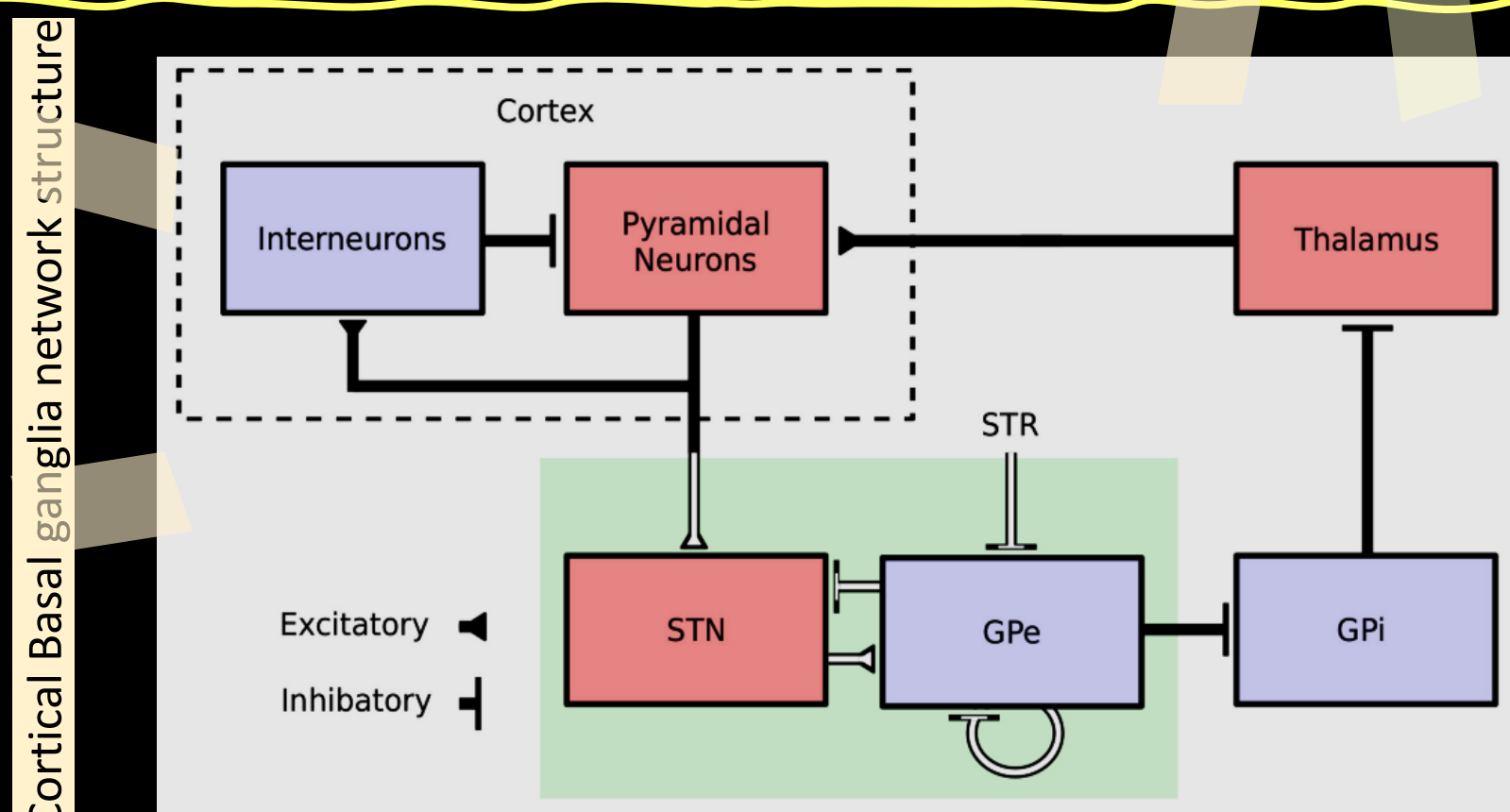
Self-tuning Deep Brain Stimulation External Globus Pallidus Observer for Suppression of Beta Oscillations

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Background & Motivation

Pathological beta oscillations are one of the major manifestations of Parkinson's disease (PD), and inhibiting their production is critical to the treatment of Parkinson's disease. It was found that changes in the frequency of neuronal firing in the subthalamic nucleus (STN) and globus pallidus (GPe) regions lead to abnormal beta oscillations. Deep brain stimulation (DBS) can be used to control this areas of the brain. Most of the closed-loop DBSs proposed and tested to date offer advantages over conventional open-loop DBSs, but the performance of the controller may be less than optimal if operating conditions change [1]. For example, daily variations in symptoms, disease progression, or changes in electrode-tissue interface properties may lead to such changes. In contrast, adaptive or "self-tuning" control mechanisms have the potential to accommodate slow changes in system properties over days, months or years.



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Aim

The effects of the frequencies generated by the STN and GPe feedback loops on pathological beta oscillations have been explored deeply, but the specifics regarding neuronal activity in the GPe region are not well understood [3]. Studying the state of neuronal activity in the GPe region by designing a suitable observer is beneficial for the further development of DBS for Parkinson's disease.

Details

This dynamical system is nonlinear and the traditional Luenberger observer is based on a linear system, so the Kazantzis Kravaris/Luenberger (KKL) [2] observer is applied. The KKL observer is based on a special coordinate transformation that estimates the system state by embedding the nonlinear system into a linear stabilising filter, which is robust but insensitive to measurement noise, and therefore needs to be combined with a neural network observer. A neural network can be used to represent the output behaviour and then the KKL observer can be used to estimate the internal state of the network, allowing for a deeper understanding and control of the network behaviour. The most commonly used models for describing neuronal activity in neuroscience and control theory are the firing rate model and the Conductance-Based Model [1]. The firing rate model is typically used to describe the average behaviour of large neural networks and is more efficient when dealing with large-scale networks but does not allow for tracking of the spiking actions of each neuron. The Conductance-Based Model is capable of describing the biophysical properties of neuronal cell membranes, and it provides a more detailed biophysical description of neuronal dynamics. A complex-valued firing rate model can be derived using the eigenfunction expansion of the Fokker-Planck equation [4], which reproduces the ignition rate dynamics due to partial synchronisation of action potentials in the spiking model, thus tracking the spiking action. Combining this model with the Conductance-Based Model gives more comprehensive results.

References

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